Rounding and estimation



Year 7 Set 3 · White Rose KS3 · 60 minutes · Mastery lesson

LO1: Round to the nearest 10, 100, 1000 and to 1 decimal place
LO2: Round to 1 or 2 significant figures
LO3: Estimate answers by rounding to 1 significant figure

Lesson for Deniz · 60 minutes · OnMaths · Mr Ali

Have ready: mini whiteboard, pen, calculator OFF

Do Now

8 minutes

On your own. Show your working. Then we check together.

L2: What is the value of the 4 in 2.341?

L2: Order 0.7, 0.07, 0.71 (smallest first)

L1: Solve x + 18 = 40

Skill: Is 46 nearer to 40 or to 50?

Talk Task

3 minutes

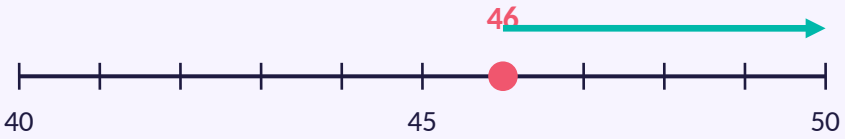
Always, Sometimes or Never?
“Rounding a number makes it smaller.”

Say: I think it is _____ because _____ . For example, _____ .

Key facts: rounding

Key facts

46 is between 40 and 50. It is nearer 50.



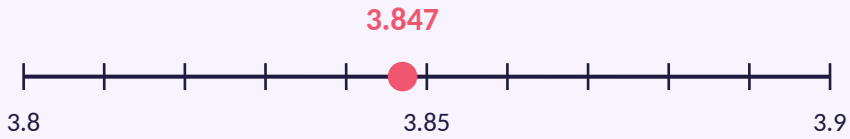
Place-holder zeros keep the size: 6 482 ≈ 6 000 (not 6)

Next digit 5 or more	round UP
Next digit 4 or less	keep the same
1st significant figure	the first digit that is not 0
Estimate	round every number to 1 s.f., then work it out
≈	means ‘approximately equal to’

LO1 I do: round to 10, 100 and 1 d.p.

I do

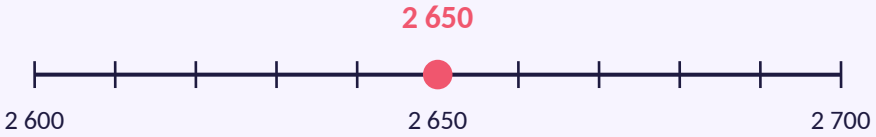
Round 3.847 to 1 decimal place.



LO1 I do: example 2

I do

Round 2 650 to the nearest 100.



LO1 We do

We do

You say it first, then click.

74 to the nearest 10

385 to the nearest 100

6.27 to 1 d.p.

14.95 to 1 d.p.

LO1 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

5 280 to the nearest 1 000

0.46 to 1 decimal place

LO1 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

19.96 to 1 decimal place

Ali rounds 7.35 to 1 d.p. and gets 7.3. What is his mistake?

LO1 You do

B · S · G · P

BRONZE

Round 63 to the nearest 10.

SILVER

Round 4.72 to 1 decimal place.

GOLD

Round 9.981 to 1 decimal place.

PLATINUM

A number rounds to 3.6 (1 d.p.). What is the smallest it could be?

LO2 I do: significant figures

I do

Round 6 482 to 1 significant figure.

Th	H	T	O
6	4	8	2

1st s.f.

LO2 I do: example 2

I do

Round 0.0573 to 1 significant figure.



LO2 We do

We do

You say it first, then click.

47 to 1 s.f.

812 to 1 s.f.

0.38 to 1 s.f.

3 849 to 2 s.f.

LO2 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

2 570 to 1 significant figure

0.0091 to 1 significant figure

LO2 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

56 400 to 2 significant figures

Sam says 4 739 to 1 s.f. is 5. What is wrong?

LO2 You do

B · S · G · P

BRONZE

Round 38 to 1 significant figure.

SILVER

Round 0.264 to 1 significant figure.

GOLD

Round 7 952 to 2 significant figures.

PLATINUM

A town has 48 650 people. Round this to 1 s.f. and to 2 s.f.

LO3 I do: estimate

I do

Estimate 39 × 21

Real answer: 819
800 is close – a good estimate

LO3 I do: example 2

I do

Estimate this calculation:

$$\frac{61 + 38}{4.9}$$

Work out the top first

LO3 We do

We do

Estimate. You say it first, then click.

52×9.8

$397 \div 8.1$

6.2×3.9

$28 + 71$

LO3 Mini whiteboards (1 of 2)

MWB

Write your answer, show me, then click.

Estimate 198 × 5.1

Estimate 0.48 × 61

LO3 Mini whiteboards (2 of 2)

MWB

Write your answer, show me, then click.

A pen costs £1.95. Estimate the cost of 31 pens.

Estimate 4.1 × 5.2. Is the real answer more or less? Why?

LO3 You do

B · S · G · P

BRONZE

Estimate 29×11

SILVER

Estimate $812 \div 39$

GOLD

Estimate:

$$\frac{19.6 \times 3.1}{0.49}$$

PLATINUM

58 students each pay £9.85. The teacher estimates £600. Is this too much or too little? Why?

Exam questions

Exam style

1. **Round 7.462 to 1 decimal place.**

(1 mark)

2. **Round 0.0386 to 1 significant figure.**

(1 mark)

3. **Work out an estimate for**

$$\frac{302 \times 9.7}{0.51}$$

(3 marks)

Exit ticket

5 minutes

On your own. Also on your OnMaths page, where it is marked for you.

1. **Round 12.68 to 1 decimal place.**

2. **Round 3 849 to 2 significant figures.**

3. **Estimate 198 × 5.1**

Finished early? Extension

Extension

Round 999.96 to 1 decimal place.

Write a number that rounds to 300 (1 s.f.) and to 250 (2 s.f.).

Estimate 3.14×9.8^2

Homework

Due next lesson

5 questions on rounding and estimation (7 marks).

Do it online on your OnMaths page – it marks itself.
Watch the 2-minute video on your page if you get stuck.
Show your working on paper.

Today I learned: round to 10, 100, 1 d.p. ✓ significant figures ✓ estimate ✓